Markov Numbers Do Not Have Square-Root Counting Growth: A Disproof of Conjecture 00000008333

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Abstract

Let $M(X)$ count the distinct positive integer Markov numbers at most X. We prove the explicit estimate

$$M(64^n) \le 2 + 3(6n+1)^2\, 2^{2n+1} \qquad (n \ge 0).$$

Consequently $M(64^n)/8^n \longrightarrow 0$, contradicting $M(X) \sim c\sqrt{X}$ for every $c > 0$. The argument uses elementary descent, a complete binary tree of actual Markov triples, and a bound on its words after their second right branch. Lean verifies the full bridge from positive integer solutions to the actual counting function and the quantified negation of the asserted asymptotic.

Reading of the source

The English source asserts square-root growth for the distribution of Markov numbers; the Chinese version explicitly writes mark_dist $\sim cx^{1/2}$. We use the usual cumulative counting interpretation

$$\mathcal{M} = \{m \in \mathbb{Z}_{>0} : \ \exists a, b \in \mathbb{Z}_{>0},\ a^2 + b^2 + m^2 = 3abm\},$$
$$M(X) = \#\{m \in \mathcal{M} : m \le X\}.$$

Numbers are counted once, even if they occur in several triples. The constant in an asymptotic growth law is positive. It suffices to consider integer cutoffs: an asserted asymptotic for all real cutoffs would also hold along the integers.

The source also discusses prime Markov numbers, Frobenius uniqueness, and random walks. We refute its square-root clause, which is necessary for the conjunction asserted there. We make no claim to settle its other clauses. In particular, the proof neither assumes nor establishes uniqueness of the largest coordinate among tree nodes.

For historical context, Zagier proved a much sharper asymptotic for the number of sorted Markov triples below a bound [1]. That count allows repeated largest-coordinate labels. We do not identify it with the count of distinct numbers or use that asymptotic in this proof. The elementary upper bound below is verified directly.

Descent and complete tree coverage

Call (a, b, c) a sorted Markov triple when $0 < a \le b \le c$ and

$$a^2 + b^2 + c^2 = 3abc. \tag{1}$$

Starting with $T_\varnothing = (1, 2, 5)$, define children of a strict sorted triple by

$$L(a, b, c) = (a, c, 3ac - b), \qquad R(a, b, c) = (b, c, 3bc - a).$$

For a finite word w in L, R, let T_w be its value, applying its letters from left to right, and let $C(w)$ be its largest coordinate.

Lemma 1. *Every sorted Markov triple is* $(1,1,1)$, $(1,1,2)$, *or some* T_w. *Every* T_w *is strict:* $0 < a < b < c$.

Proof. The substitution $b \mapsto 3ac - b$ preserves (1); this follows by expansion. For a strict triple, $a \geq 1$ and $b < c$, so $3ac - b > c$. The same argument gives $3bc - a > c$. Thus both children are strict Markov triples, and the root is one as well.

For completeness, take a sorted solution other than one with $a = b = 1$, and set $d = 3ab - c$. First $c < 3ab$: the opposite inequality would give $3abc \leq c^2$, contradicting (1) and $a, b > 0$. Hence $d > 0$, and

$$d + c = 3ab, \qquad dc = a^2 + b^2.$$

We claim $d < b$. Since $(a, b) \neq (1, 1)$,

$$a^2 + 2b^2 < 3ab^2. \tag{2}$$

Indeed, for $a = 1$ we have $b \geq 2$; for $a \geq 2$ we use $a^2 \leq b^2$ and $3ab^2 \geq 6b^2 > 3b^2$. If $d \geq b$, then $(d - b)(c - b) \geq 0$, whereas the displayed identities give

$$(d - b)(c - b) = a^2 + 2b^2 - 3ab^2 < 0.$$

This is a contradiction. Also $b < c$, because $b = c$ would imply $db = a^2 + b^2 > b^2$ and hence $d > b$.

Mutation shows that (a, b, d) again satisfies the equation. If $d \leq a$, its sorted version is (d, a, b), and the original triple is its R child. If $d > a$, its sorted version is (a, d, b), and the original is its L child. Either parent has smaller coordinate sum. Induction therefore reduces to the case $a = b = 1$, where the equation gives $c = 1$ or 2. The parent $(1, 1, 1)$ can lead only to $(1, 1, 2)$; the first strict triple after $(1, 1, 2)$ is $(1, 2, 5)$. These are exactly the exceptional triples and root stated above. □

Lemma 2. *For every* $m \in \mathcal{M}$, *either* $m = 1$, $m = 2$, *or* $C(w) = m$ *for some finite word* w.

Proof. Sort any positive solution containing m and use Lemma 1. At the root, its coordinates are $1, 2, 5$, and $5 = C(\varnothing)$. Each child reuses two coordinates from its parent and creates just its own new largest coordinate. Induction along its path therefore shows that every coordinate is either 1, 2, or the largest coordinate of an ancestor. This argument does not bound the largest coordinate of an arbitrary triple containing a small m: it chooses the ancestor at which m is a largest coordinate. □

A bound on words below a cutoff

For a strict triple (a, b, c), either child's largest coordinate is at least $(3a-1)c$. For L this follows from $b \leq c$; for R it follows from $b \geq a$ and $a \leq c$. A child never decreases the least coordinate. In addition, every R increases that coordinate by at least one, since it replaces a by $b > a$. It follows that

1. $C(w) \geq 2^{|w|} \cdot 5$, so $C(w) \leq 64^n$ implies $|w| \leq 6n$;
2. after two R letters the least coordinate is at least three, so each remaining step increases the largest coordinate by a factor of at least eight. If $w = uv$, u contains at least two R letters, and $C(w) \leq 64^n$, then $8^{|v|} \leq 64^n$ and $|v| \leq 2n$.

These statements follow equally for any initial strict triple by induction, multiplying its largest coordinate by the stated factor at each step.

Write $A = 6n + 1$. Every word with $C(w) \leq 64^n$ belongs to one of three classes, according to its first two R letters:

$$L^i, \qquad L^iRL^j, \qquad L^iRL^jRv.$$

Here $0 \le i, j \le 6n$, and in the third class $|v| \le 2n$. The first two classes explicitly include paths stopping before the second R. There are at most A choices of the first type, A^2 of the second, and

$$A^2 \sum_{k=0}^{2n} 2^k \le A^2\, 2^{2n+1}$$

of the third. Thus the total is at most $3A^2\, 2^{2n+1}$. We need only an upper bound: different words may have the same largest coordinate.

By Lemma 2, every distinct $m \in \mathcal{M}$ at most 64^n, except possibly $1, 2$, is the largest coordinate of a word in this finite collection. Taking its image under C gives

$$M(64^n) \le 2 + 3(6n+1)^2\, 2^{2n+1}. \tag{3}$$

This is a bound on the actual set of positive integer Markov numbers, not just a count of an independently specified tree.

Contradiction to square-root growth

Dividing (3) by $8^n = \sqrt{64^n}$ gives

$$0 \le \frac{M(64^n)}{8^n} \le \frac{2}{8^n} + 216\frac{n^2}{2^n} + 72\frac{n}{2^n} + 6\frac{1}{2^n} \longrightarrow 0.$$

Each term tends to zero because a fixed polynomial grows more slowly than a geometric sequence of ratio greater than one. In particular, for every $c > 0$ and every N there is $n \ge N$ with $M(64^n) < c8^n$.

Theorem 1. *There is no real $c > 0$ for which $M(X)/\sqrt{X} \longrightarrow c$ as the integer cutoff $X \longrightarrow \infty$.*

Proof. Such a limit, restricted to $X = 64^n \longrightarrow \infty$, would give $M(64^n)/8^n \longrightarrow c$. The same sequence tends to zero, so uniqueness of limits would give $c = 0$, a contradiction. Thus the square-root clause, and hence the source's conjunction, is false under the stated counting interpretation. □

Formal verification and scope

The Lean development defines the actual equation over natural numbers with all three coordinates positive. Its finite set at cutoff X filters $\{0, \dots, X\}$ by existence of positive a, b satisfying the equation; its cardinal is exactly $M(X)$. No table of Markov numbers is used. The descent, sorted tree coverage, six sorting cases, coordinate birth lemma, path growth, three word classes, finite-set image bound, real limit, and final negation are all formally proved.

The principal statements, in namespace `Markov8333`, are

`number_birth`, `count_bound`,
`count_subsequence_zero`, `arbitrarily_small_ratio`,
`conjecture_8333_false` : ¬ `SquareRootGrowth`.

Here `SquareRootGrowth` is the existential positive-constant limit in the theorem above, with the actual finite-set cardinal. Lean also proves $\sqrt{64^n} = 8^n$. The final theorem is not a conditional statement assuming completeness of the tree or uniqueness of Markov labels. The interpretation of the source's word "distribution" as a cumulative count, and the observation that a real-cutoff asymptotic implies its integer restriction, are stated explicitly here. No assertion about the other three source clauses is formalized or needed.

The Lean source contains no unproved placeholders or additional axioms. The reported axioms of the five audited theorems are only `propext`, `Classical.choice`, and `Quot.sound`. Development and self-review were performed by a delegated AI agent; the coordinating agent reviews separately. No external independent review is claimed.

Source and reproduction

The exact bilingual statement is archived as `SOURCE.md`, from the competition repository at commit

`b355e42906e6eaf52bc7e9d005e0ef02d8aa7400`.

The source path is

`conjectures/00000008333.md`.

Source hashes and snapshot checks are recorded in the accompanying provenance file. No earlier submission was found in that snapshot.

The project pins Lean 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7709c5d26b00b`.

From the supplied `lean` directory, run

```
lake env lean -DwarningAsError=true Main.lean
```

The proof is self-contained apart from standard finite-set, arithmetic, and real-analysis facts supplied by Mathlib.

References

[1] D. Zagier, *On the Number of Markoff Numbers Below a Given Bound*, Mathematics of Computation 39 (160), 1982, 709–723. Author's public copy.